



Qualification Pack



Fundamental for IoT for Home Automation

QP Code: SVO/ELE/Q0201

Version: 1.0

NSQF Level: 4

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SVO/ELE/Q0201: Fundamental for IoT for Home Automation

Brief Job Description

This unit is about performing installation, onsite troubleshooting and repairing of the home automation systems as per the specifications of the manufacturer and consultation with the customer addressing his/her needs

Personal Attributes

Process of Installation, Troubleshooting and Repairing of IoT - Home Automation Systems

Applicable National Occupational Standards (NOS)

Compulsory NOS:

1. [SVO/ELE/N0201: Fundamental for IoT for Home Automation](#)

Qualification Pack (QP) Parameters

Sector	Electronics
Sub-Sector	Consumer Electronics
Occupation	Fundamental for IoT for Home Automation
Country	India
NSQF Level	4
Credits	4
Aligned to NCO/ISCO/ISIC Code	NCO-2015/2144.0801
Minimum Educational Qualification & Experience	10th Class + I.T.I with NA of experience OR 12th Class with NA of experience OR Completed 2nd year of the 3-year diploma after 10 with NA of experience
Minimum Level of Education for Training in School	10th Class



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Pre-Requisite License or Training	NA
Minimum Job Entry Age	15 Years
Last Reviewed On	NA
Next Review Date	30/04/2027
NSQC Approval Date	30/04/2024
Version	1.0
Reference code on NQR	NG-04-EH-02532-2024-V1-SCTEVTO
NQR Version	1



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SVO/ELE/N0201: Fundamental for IoT for Home Automation

Description

This unit is about performing installation, onsite troubleshooting and repairing of the home automation systems as per the specifications of the manufacturer and consultation with the customer addressing his/her needs.

Scope

The scope covers the following :

- Preparatory activities for installation of home automation system
- Installation of smart devices and sensors at home
- Troubleshooting and repairing of defects in the smart devices

Elements and Performance Criteria

To be competent, the user/individual on the job must be able to:

- PC1.**
 - Assess and identify the customer requirements for the installation of automated devices (e.g. smart lighting, smart security system for doors, smart plugs, smart entertainment system, smart thermostat, smart sensors for garden, smart sensors for toilet etc.) at customer home
- PC2.** Collect the necessary tools and equipment required for installation and repairing work
- PC3.** Check the functionality of tools and equipment before use
- PC4.**
 - Select the center hub or controller such as customer smartphone to control the smart devices as per the customer preference
- PC5.**
 - Select and arrange the required wires, cables, sensors, lighting system and automated systems need to install
- PC6.**
 - Check the smart devices specifications and ensure that all supporting accessories purchased are available in the pack
- PC7.**
 - Check the availability of wi-fi connection at the customer premises and ensure that its speed is enough for running all the smart devices smoothly and properly
- PC8.**
 - Plan for the installation process by considering factors such as device placement, wiring requirements, and network connectivity
- PC9.** Use appropriate personal protective equipment (PPE) for specific tasks and work conditions
- PC10.** Coordinate with the relevant personnel in case of any issues in the installation
- PC11.** Determine and mark the ideal locations for fixing or placing the smart devices and sensors
- PC12.**
 - Ensure nearest access to power outlets and internet connectivity for connecting the smart devices and sensors



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- PC13.** • Take measurements and fix or route the wires in customer premises for connecting the smart devices
• and sensors with the power plugs
- PC14.** • Fix lighting systems in the customer premises on the marked places in customer premises by following
• the SOP and manufacturer manual
- PC15.** • Fix smart door bell, video camera and door lock at the door by following the SOP and manufacturer
• manual
- PC16.** • Fix CCTV cameras and other surveillance systems for home by following the SOP and manufacturer
• manual
- PC17.** • Fix smart systems and sensors in kitchen, bathroom, garden etc. as per the customer requirement by
• following the SOP and manufacturer manual
- PC18.** Connect all systems and sensors in the power plug and switch on the power
- PC19.** • Install the smart devices applications in the center hub or customer smartphone and connect the
• smart devices with customer's phone
- PC20.** • Configure the settings, preferences, and automation routines according to customer requirements in
• the smart application
- PC21.** • Test each device and system to ensure proper functionality and verify that all devices respond
• correctly to commands, test automation scenarios, and troubleshoot issues after the installation
- PC22.** • Demonstrate and explain the various functions, features and settings of smart devices and use of
• control panel to the customer
- PC23.** Demonstrate how to operate and monitor the functioning of smart devices through application
- PC24.** • Guide the customer on day-to-day care activities and precautions while using the smart devices as
• mentioned in product manual
- PC25.** prepare and maintain the records and documents related to installation work done
- PC26.** Enquire about the problems/unusual conditions noticed on the smart devices from the customer
- PC27.** • Use the smart feature app installed in customer's smartphone to diagnose the issues in the smart
• devices by following instructions in manufacturer manual
- PC28.** Carry out troubleshooting activities to identify the faults and faulty component in the smart devices
- PC29.** • Unplug the unit, carry out basic tests such as power supply inspection, volt ampere test and earth test
• power supply etc. as per the instructions in manufacturer manual and SOP
- PC30.** • Remove the faulty component for carrying out repair or replacement after disconnecting it from
• power supply



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- PC31.** • Repair the fault at customer premises if possible or send the device to service center for repairing or
• replacement, if the problem cannot be serviced onsite
- PC32.** • Test the functionality of device and system and verify that it is now responding correctly to
• commands and test automation scenarios after repairing
- PC33.** Prepare and record the details related to troubleshooting and repairing of device

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** Relevant legislation, standards, policies, and procedures followed in the organization
- KU2.** Health and safety requirements applicable in the workplace
- KU3.** • Various types of health and safety hazards commonly present in the work environment such as physical
• hazards, electrical hazards, chemical hazards, fire hazards, equipment related hazards, health hazards, etc.
- KU4.** Importance of using protective clothing/equipment while working
- KU5.** Main hazards and preventive as well as control measures while working with different types of equipment
- KU6.** • Importance of carrying out electrical and non-electrical isolation to prevent hazards from loss of
• machine/system/process control
- KU7.** • Main hazards and preventive as well as control measures when working with electrical systems or using
• electrical equipment
- KU8.** Fundamentals of electricity such as ohms law, difference between ac and dc
- KU9.** • Basic electronics (knowledge of components such as diode, transformer, LED, photo transistor, capacitor,
• resistor, inductor, thermistor, ICs etc.)
- KU10.** Working technology of IoT devices
- KU11.** Various components of IoT systems
- KU12.** IoT protocols for smart home
- KU13.** Benefits of smart home systems
- KU14.** Application of smart devices in different areas of a home
- KU15.** • Functioning, use and features of various automated devices and sensors i.e. smart lighting, smart security
• system for doors, smart plugs, smart entertainment system, smart thermostat, smart sensors for garden,
• smart sensors for toilet etc. for home
- KU16.** Selection criteria of home automated devices
- KU17.** • Requirements i.e. controller, platform, connectivity, sensors, devices etc. for installation of various
• automated devices and sensors
- KU18.** Installation site requirements (structural requirements, power, etc.)
- KU19.** Tools and equipment required during installation, troubleshooting and repairing activities



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- KU20.** Procedure of setup and installation of various automated devices and sensors at home
- KU21.** Installation, configuration and use of smart system applications
- KU22.** How to operate and control smart systems
- KU23.** Various issues and their causes in smart devices
- KU24.** Troubleshooting and repairing activities for various automated devices and sensors
- KU25.** Reports and documents need to fill and maintain related to installation and repairing work

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** Follow instructions, guidelines, procedures, rules, and service level agreements
- GS2.**
 - Read, interpret and follow information on written job instructions, specifications, standard operating
 - procedures, and other applicable reference documents
- GS3.** Planning and sequencing of installation activities
- GS4.** Checking and clarifying task related information
- GS5.** Discuss task lists, schedules and activities with the seniors and team members
- GS6.** Follow organization rule-based decision making process
- GS7.** Take decisions with systematic course of actions and/or response
- GS8.** Plan and organize tasks to meet deadlines
- GS9.** Find ways of modifying difficult operating stages to make it operation friendly
- GS10.**
 - Apply domain information to set and define operation parameters that ensures economy and quality of
 - the product
- GS11.**
 - Analyze the complexity of work to determine if it can be successfully carried out or needs to be referred
 - to a superior/specialist
- GS12.** Recognize a workplace problem and take suitable action to resolve it



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Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
	30	50	-	20
PC1. - Assess and identify the customer requirements for the installation of automated devices (e.g. smart lighting, smart security system for doors, smart plugs, smart entertainment system, smart thermostat, - smart sensors for garden, smart sensors for toilet etc.) at customer home	1	1	-	1
PC2. Collect the necessary tools and equipment required for installation and repairing work	1	2	-	1
PC3. Check the functionality of tools and equipment before use	1	1	-	-
PC4. - Select the center hub or controller such as customer smartphone to control the smart devices as per - the customer preference	1	1	-	1
PC5. - Select and arrange the required wires, cables, sensors, lighting system and automated systems need - to install	2	2	-	1
PC6. - Check the smart devices specifications and ensure that all supporting accessories purchased are - available in the pack	1	1	-	-
PC7. - Check the availability of wi-fi connection at the customer premises and ensure that its speed is - enough for running all the smart devices smoothly and properly	1	1	-	-
PC8. - Plan for the installation process by considering factors such as device placement, wiring - requirements, and network connectivity	1	2	-	1



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Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC9. Use appropriate personal protective equipment (PPE) for specific tasks and work conditions	-	1	-	-
PC10. Coordinate with the relevant personnel incase of any issues in the installation	-	1	-	-
PC11. Determine and mark the ideal locations for fixing or placing the smart devices and sensors	1	1	-	1
PC12. • Ensure nearest access to power outlets and internet connectivity for connecting the smart devices • and sensors	-	1	-	-
PC13. • Take measurements and fix or route the wires in customer premises for connecting the smart devices • and sensors with the power plugs	1	2	-	1
PC14. • Fix lighting systems in the customer premises on the marked places in customer premises by following • the SOP and manufacturer manual	1	2	-	1
PC15. • Fix smart door bell, video camera and door lock at the door by following the SOP and manufacturer • manual	1	2	-	1
PC16. • Fix CCTV cameras and other surveillance systems for home by following the SOP and manufacturer • manual	1	2	-	1
PC17. • Fix smart systems and sensors in kitchen, bathroom, garden etc. as per the customer requirement by • following the SOP and manufacturer manual	1	2	-	1
PC18. Connect all systems and sensors in the power plug and switch on the power	-	1	-	-



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Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC19. - Install the smart devices applications in the center hub or customer smartphone and connect the - smart devices with customer's phone	1	2	-	1
PC20. - Configure the settings, preferences, and automation routines according to customer requirements in - the smart application	1	2	-	-
PC21. - Test each device and system to ensure proper functionality and verify that all devices respond - correctly to commands, test automation scenarios, and troubleshoot issues after the installation	1	2	-	1
PC22. - Demonstrate and explain the various functions, features and settings of smart devices and use of - control panel to the customer	1	1	-	1
PC23. Demonstrate how to operate and monitor the functioning of smart devices through application	1	1	-	-
PC24. - Guide the customer on day-to-day care activities and precautions while using the smart devices as - mentioned in product manual	1	1	-	-
PC25. prepare and maintain the records and documents related to installation work done	1	1	-	-
PC26. Enquire about the problems/unusual conditions noticed on the smart devices from the customer	1	1	-	1
PC27. - Use the smart feature app installed in customer's smartphone to diagnose the issues in the smart - devices by following instructions in manufacturer manual	1	1	-	1
PC28. Carry out troubleshooting activities to identify the faults and faulty component in the smart devices	2	3	-	1



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Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC29. - Unplug the unit, carry out basic tests such as power supply inspection, volt ampere test and earth test - power supply etc. as per the instructions in manufacturer manual and SOP	1	2	-	1
PC30. Remove the faulty component for carrying out repair or replacement after disconnecting it from power supply	1	2	-	1
PC31. Repair the fault at customer premises if possible or send the device to service center for repairing or replacement, if the problem cannot be serviced onsite	1	2	-	1
PC32. Test the functionality of device and system and verify that it is now responding correctly to commands and test automation scenarios after repairing	1	2	-	-
PC33. Prepare and record the details related to troubleshooting and repairing of device	-	1	-	-
NOS Total	30	50	-	20



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National Occupational Standards (NOS) Parameters

NOS Code	SVO/ELE/N0201
NOS Name	Fundamental for IoT for Home Automation
Sector	Electronics
Sub-Sector	
Occupation	Fundamental for IoT for Home Automation
NSQF Level	4
Credits	4
Version	1.0
Last Reviewed Date	30/04/2024
Next Review Date	30/04/2027
NSQC Clearance Date	30/04/2024

Assessment Guidelines and Assessment Weightage

Assessment Guidelines

1. Assessment System Overview:
 - Batches assigned to the assessment agencies for conducting the assessment on SDMS/SIP or email
 - Assessment agencies send the assessment confirmation to TP
 - Assessment agency deploys the ToA certified Assessor for executing the assessment
 -



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Training agency monitors the assessment process & records

2.

Testing Environment:

-

Confirm that the centre is available at the same address as mentioned on SDMS or SIP

-

Check the duration of the training.

-

Check the Assessment Start and End time to be as 10 a.m. and 5 p.m.

-

If the batch size is more than 30, then there should be 2 Assessors.

-

Check that the allotted time to the candidates to complete Theory & Practical Assessment is correct.

-

Check the mode of assessment—Online (TAB/Computer) or Offline (OMR/PP).

-

Confirm the number of TABs on the ground are correct to execute the Assessment smoothly.

-

Check the availability of the Lab Equipment for the particular Job Role.

3.

Assessment Quality Assurance levels / Framework:

-

Question papers created by the Subject Matter Experts (SME)

-

Question papers created by the SME verified by the other subject Matter Experts

-



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Questions are mapped with NOS and PC

-

Question papers are prepared considering that level 1 to 3 are for the unskilled & semi-skilled individuals, and level 4 and above are for the skilled, supervisor & higher management

-

Assessor must be ToA certified & trainer must be ToT Certified

-

Assessment agency must follow the assessment guidelines to conduct the assessment

4.

Types of evidence or evidence-gathering protocol:

-

Time-stamped & geotagged reporting of the assessor from assessment location

-

Centre photographs with signboards and scheme specific branding

-

Biometric or manual attendance sheet (stamped by TP) of the trainees during the training period

-

Time-stamped & geotagged assessment (Theory + Viva + Practical) photographs & videos

5.

Method of verification or validation:

-

Surprise visit to the assessment location

-

Random audit of the batch

-

Random audit of any candidate



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6.

Method for assessment documentation, archiving, and access

-

Hard copies of the documents are stored

-

Soft copies of the documents & photographs of the assessment are uploaded / accessed from Cloud Storage

-

Soft copies of the documents & photographs of the assessment are stored in the Hard Drives

Minimum Aggregate Passing % at QP Level : 70

(Please note: Every Trainee should score a minimum aggregate passing percentage as specified above, to successfully clear the Qualification Pack assessment.)

Assessment Weightage

Compulsory NOS

National Occupational Standards	Theory Marks	Practical Marks	Project Marks	Viva Marks	Total Marks	Weightage
SVO/ELE/N0201.Fundamental for IoT for Home Automation	30	50	-	20	100	100
Total	30	50	-	20	100	100



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Acronyms

NOS	National Occupational Standard(s)
NSQF	National Skills Qualifications Framework
QP	Qualifications Pack
TVET	Technical and Vocational Education and Training



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Glossary

Sector	Sector is a conglomeration of different business operations having similar business and interests. It may also be defined as a distinct subset of the economy whose components share similar characteristics and interests.
Sub-sector	Sub-sector is derived from a further breakdown based on the characteristics and interests of its components.
Occupation	Occupation is a set of job roles, which perform similar/ related set of functions in an industry.
Job role	Job role defines a unique set of functions that together form a unique employment opportunity in an organisation.
Occupational Standards (OS)	OS specify the standards of performance an individual must achieve when carrying out a function in the workplace, together with the Knowledge and Understanding (KU) they need to meet that standard consistently. Occupational Standards are applicable both in the Indian and global contexts.
Performance Criteria (PC)	Performance Criteria (PC) are statements that together specify the standard of performance required when carrying out a task.
National Occupational Standards (NOS)	NOS are occupational standards which apply uniquely in the Indian context.
Qualifications Pack (QP)	QP comprises the set of OS, together with the educational, training and other criteria required to perform a job role. A QP is assigned a unique qualifications pack code.
Unit Code	Unit code is a unique identifier for an Occupational Standard, which is denoted by an 'N'
Unit Title	Unit title gives a clear overall statement about what the incumbent should be able to do.
Description	Description gives a short summary of the unit content. This would be helpful to anyone searching on a database to verify that this is the appropriate OS they are looking for.
Scope	Scope is a set of statements specifying the range of variables that an individual may have to deal with in carrying out the function which have a critical impact on quality of performance required.



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Knowledge and Understanding (KU)	Knowledge and Understanding (KU) are statements which together specify the technical, generic, professional and organisational specific knowledge that an individual needs in order to perform to the required standard.
Organisational Context	Organisational context includes the way the organisation is structured and how it operates, including the extent of operative knowledge managers have of their relevant areas of responsibility.
Technical Knowledge	Technical knowledge is the specific knowledge needed to accomplish specific designated responsibilities.
Core Skills/ Generic Skills (GS)	Core skills or Generic Skills (GS) are a group of skills that are the key to learning and working in today's world. These skills are typically needed in any work environment in today's world. These skills are typically needed in any work environment. In the context of the OS, these include communication related skills that are applicable to most job roles.
Electives	Electives are NOS/set of NOS that are identified by the sector as contributive to specialization in a job role. There may be multiple electives within a QP for each specialized job role. Trainees must select at least one elective for the successful completion of a QP with Electives.
Options	Options are NOS/set of NOS that are identified by the sector as additional skills. There may be multiple options within a QP. It is not mandatory to select any of the options to complete a QP with Options.